Example Co.

Process Capability Study

27 Oct 2026

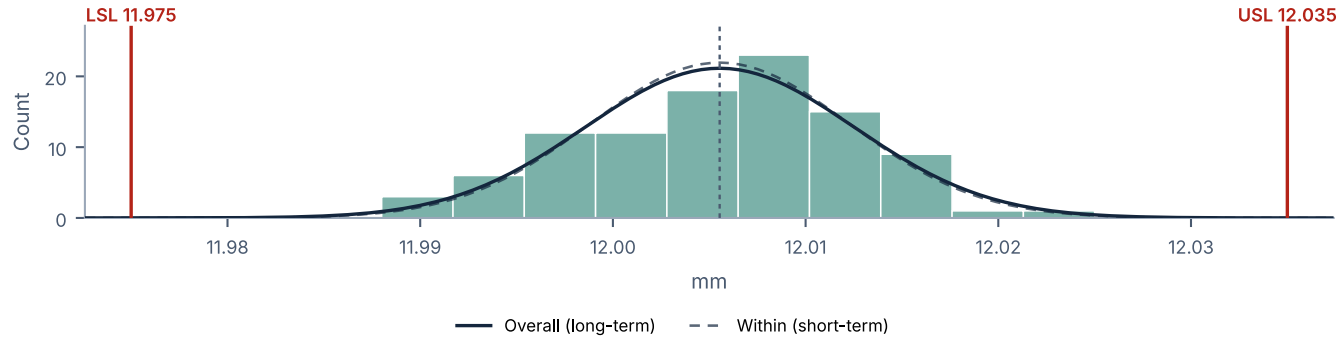
PART NUMBER **SAMPLE-002**	CHARACTERISTIC **Bore diameter (simulated data)**	SPECIFICATION **LSL 11.975 · USL 12.035 mm**
GAGE Not given	SAMPLE **100 readings · 20 subgroups of 5**	CUSTOMER Not given

CONDITIONALLY ACCEPTABLE

Ppk is 1.41, between 1.33 and 1.67. The process may be acceptable with customer approval.

PPK	PP	CPK	CP
1.41	**1.43**	**1.46**	**1.49**
Overall · verdict index	Overall	Within	Within

Distribution against the specification



	Within	Overall
Standard deviation	0.00673	0.00698
Cp / Pp	1.49	1.43
Cpk / Ppk	**1.46**	**1.41**
Cpu / Ppu	1.46	1.41
Cpl / Ppl	1.51	1.46
Expected PPM below LSL	2.8	6.1
Expected PPM above USL	6.0	12.2
Expected PPM total	**8.8**	**18.3**

Mean	**12.0055 mm**
Readings	**100 readings · 20 subgroups of 5**
Min / max	**11.988 / 12.025**
Observed PPM	**0 (0 of 100 outside)**

Prepared by · Sample report

Approved by

Date

Stability

20 subgroups of 5

Capability numbers only predict future parts if the process was stable while the data was collected. Limits are $\bar{\bar{x}} \pm A_2\bar{R}$ for averages and $D_3\bar{R}$ to $D_4\bar{R}$ for ranges (A_2 = 0.577, D_4 = 2.114).

Average (x̄) chart

Center $\bar{\bar{x}}$ = 12.006, UCL = 12.01, LCL = 12.00.



Range (R) chart

Center $\bar{R}$ = 0.01565, UCL = 0.03308.



- Every subgroup average and range is inside the control limits, so the process looks stable over this study.

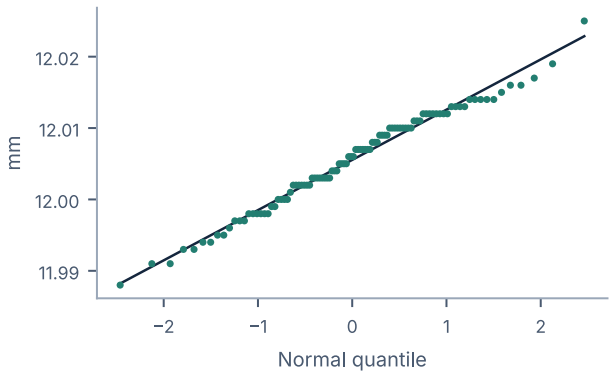
Normality

Expected PPM and the indices assume the readings follow a normal distribution. The Anderson-Darling test checks that assumption: A^2 = 0.583, p = 0.126.

- Anderson-Darling p = 0.126, so the data is consistent with a normal distribution.

Points close to the line on the probability plot also suggest normal data; a curve or S-shape suggests skew or a mixture.

Normal probability plot



Method and decision rules

How every number in this report was made

Method

Capability for a normal distribution (Montgomery, Introduction to Statistical Quality Control, 6th ed., Sections 6.2 and 8.3; NIST/SEMATECH e-Handbook 6.1.6).

Within σ = R-bar / d2 (d2 = 2.326, subgroups of 5)
Overall σ = s, the sample standard deviation (n − 1)

$Cp = (USL - LSL) / 6\sigma_{within}$ $Pp = (USL - LSL) / 6s$
$Cpk = \min(USL - \bar{x}, \bar{x} - LSL) / 3\sigma_{within}$
$Ppk = \min(USL - \bar{x}, \bar{x} - LSL) / 3s$

Expected PPM = $[\Phi((LSL - \bar{x}) / \sigma) + 1 - \Phi((USL - \bar{x}) / \sigma)] \times 10^6$
Observed PPM = readings outside the limits / $n \times 10^6$

Normality: Anderson-Darling with the D'Agostino-Stephens small-sample adjustment (statsmodels), run with 8 or more readings. Normal probabilities from SciPy.

Decision rules

PPAP-style initial process study criteria (AIAG PPAP 4th ed.), applied to Ppk.

- **Above 1.67**: acceptable.
- **1.33 to 1.67**: may be acceptable with customer approval.
- **Below 1.33**: not acceptable.

Your customer's requirements apply if they differ.

Receipt

Method	process capability (normal) (Montgomery, Introduction to SQC; NIST/SEMATECH e-Handbook 6.1.6)
Sample	20 subgroups, 5 readings per subgroup
LSL	11.975
USL	12.035
Within sigma	R-bar / d2 (d2 = 2.326, subgroups of 5)
Overall sigma	sample standard deviation (n - 1)
Verdict index	Ppk
Constant d2	2.326
Anderson-Darling	$A^2 = 0.583$, p = 0.126
Stability chart	x-bar and R control charts (Shewhart 3-sigma limits)
Engine	Mensuro engine 0.1.0
Generated	2026-10-27

How the math is checked

Before release, Mensuro's capability engine must reproduce published reference results: the NIST/SEMATECH e-Handbook worked example and reject-rate table, and Montgomery's Examples 6.1 and 8.1–8.2. The control chart limits reproduce Montgomery Example 6.1. These checks rerun on every change.

Data

All 100 readings, in mm, in the order collected

Subgroup	1	2	3	4	5	Average	Range
1	12.009	11.997	12.013	12.014	11.988	12.0042	0.026
2	11.994	12.007	12.003	12.006	11.998	12.0016	0.013
3	12.014	12.013	12.007	12.016	12.010	12.0120	0.009
4	11.998	12.009	11.997	12.014	12.006	12.0048	0.017
5	12.004	12.000	12.017	12.005	12.002	12.0056	0.017
6	12.003	12.011	12.009	12.010	12.010	12.0086	0.008
7	12.025	12.002	12.001	11.999	12.012	12.0078	0.026
8	12.016	12.005	11.998	11.999	12.012	12.0060	0.018
9	12.013	12.011	12.000	12.008	12.007	12.0078	0.013
10	12.008	12.014	12.008	12.012	12.007	12.0098	0.007
11	12.009	12.012	11.993	12.003	12.002	12.0038	0.019
12	12.000	12.004	12.019	11.998	12.015	12.0072	0.021
13	11.991	12.003	12.007	12.011	12.012	12.0048	0.021
14	12.013	12.003	12.002	12.014	12.004	12.0072	0.012
15	11.995	11.996	11.998	12.010	12.007	12.0012	0.015
16	12.012	12.002	12.007	12.012	12.003	12.0072	0.010
17	12.010	12.000	12.003	12.003	11.995	12.0022	0.015
18	12.010	12.002	12.006	12.010	12.010	12.0076	0.008
19	12.012	12.005	12.002	12.005	11.991	12.0030	0.021
20	11.993	11.994	11.997	12.010	11.998	11.9984	0.017